

Empirical Methods in Finance

Oil Shocks, the Yield Curve, and U.S. Equity Returns: Evidence from Weekly Data and Out-of-Sample Forecasting

Report 1

Matteo Piras (20806717)
Marco D'Amico (25416579)
Roberto Berardi (25419094)

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Abstract

This project aims to investigate whether oil price shocks carry useful data that can help analysts in forecasting weekly U.S. equity returns. This forecasting model is computed after taking into account the transmission mechanism of these effects through the Treasury yield curve. With over 30 years of data (January 1990 - March 2026), this report examines the correlation between oil, the yield curve and the equity market. This is done by analyzing WTI crude oil returns, the S&P 500 returns and the spread between the 10 year and 2 year treasury yield. To answer this question, this report estimates a simple OLS baseline, a dynamic ARX(1) specification, a more complex VAR(1) system and benchmark them with an AR(1) and a random walk procedure via a rolling out-of-sample forecasting exercise. The in-sample study discloses that the yield curve spread is a significant (1% level) predictor for equity returns. This significance is lost when return dynamic are controlled with the view that oil shocks affect directly the equity market rather than indirectly via the monetary policy implementations expectations. The transmission chain proposed by Bernanke et al. (1997) is supported by a VAR response, which confirms that equity responses to oil price shocks are not statistically different from zero. When using an out-of-sample perspective, the random walk model outperforms all other specifications over 1'731 weeks of forecasts. In this analysis the project found the ARX(1) and the VAR(1) to have root mean squared deviation 3% higher than the benchmark. Rolling coefficient computations show that the oil-equity relationship depend on macro-financial episodes (2008 GFC, Covid-19,...), with its sign alternating across crisis shocks, demand booms and even supply disruptions. These kinds of instabilities prevent a reliable forecasting exercise using the oil channel, which is consistent with Kilian (2009) and a weak form of the famous Efficient Market Hypothesis.

Keywords: Oil price shocks; Treasury yield spread; U.S. equity returns; Out-of-sample forecasting; Time-series models

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1 Introduction

1.1 Research Question

This report wants to answer whether oil price shocks bring useful information for U.S. equity returns after taking into account the transmission through the Treasury Yield curve. The analysis asks if weekly changes in WTI crude oil prices can help explain movements in the S&P 500 directly or whether these movements are more related to changes in the slope of the term structure. To do that, this project computes the difference between the 10-year and the 2-year Treasury spread, including its weekly variation. Therefore, the focus is on an oil–yield curve–equity mechanism rather than a simple bivariate relation between oil shocks and stock returns.

A second related question concerns forecasting performance. The underlying economic mechanism may appear plausible but this does not mean that more accurate models will yield better predictions. For that reason, the report also questions if auto-regressive structure and more advanced macro-financial variables can improve short-horizon forecasts relative to simpler benchmarks. In practice, the project computes and compares an auto-regressive model with exogenous variable (ARX(1)) and a Vector Auto-regression (VAR(1)) against an Auto-regression (AR(1)) and a Random Walk benchmark. The project does not presume that the more sophisticated the model is, the better the results are. Instead, the objective is to test whether the additional information carried by oil and yield curve can better translate into measurable forecasting value.

Combined, these questions give the project a structured core, combining empirical asset pricing, macro-finance and forecasting. The report examines whether oil shocks are relevant for equity markets, if yes through which channel they matter, and whether that information is strong enough to survive an out-of-sample forecasting exercise based on weekly financial data.

1.2 Economic Motivation

Oil remains a central and important input in production, transportation and more generally in household consumption. Cost Increases caused by oil price shocks can raise firms' costs, diminish real disposable income and modify inflation dynamics. For this reason, macroeconomic literature has persistently treated oil shocks as source of general and aggregate market's fluctuations (Hamilton, 1983). At the same time, more recent researches demonstrate that oil price shocks are not homogeneous. For example, an increase in oil prices may exhibit supply disruptions, global demand shrink or growth or all oil-market related demand pressures. These different shocks do not need to have the same implications for macroeconomic and financial variables (Kilian, 2009). This implies that, for equity markets for example, oil should not be considered as a reliable predictor of stock returns. Instead, the mechanism is more on a macro-financial level where effects depend on the transmission mechanism and on the nature and amplitude of the underlying shock.

This report uses the yield curve as a central transmission mechanism. If oil prices rise after a shock, equity markets tend to revise upward their inflation's expectations. Moreover, the yield curve may flatten if short-term rates are awaited to react quicker than long-term rates. This mechanism is consistent with the shared belief that the economic effects of oil shocks are partly mediated through endogenous policy response, and not by oil alone (Bernanke et al., 1997). This also gives us a neat role to the slope of the term structure, playing the role of a summary measure of changing financial conditions.

It has long been shown that the yield curve is particularly relevant because it contains information about the cyclical life of economic activity. A steepening curve is generally associated with a wider economic activity, whereas a slowing growth is usually anticipated by a flat or an inverted curve (Estrella & Hardouvelis, 1991; Estrella & Mishkin, 1996). This has an impact for example for equity pricing because fore-casted and expected growth, discount rates, inflation

expectations are linked together to the performance of equity market. Moreover, asset-pricing literature and studies demonstrate that stock and even bond market respond to revision in what agents expect about future cash flows, inflation and expected returns. The yield curve makes itself a strong candidate variable capable of linking macro-economic shocks, like the oil one, to equity performance (Campbell & Ammer, 1993).

This vision is supported by the literature on oil shocks and stock market. Numerical evidence does not endorse a single stable effect of oil on equities. In fact, the stock market response is usually linked to the type of oil shock and on the actual macroeconomic situation (Kilian & Park, 2009). Instead of assuming an unique and single oil effect, the report studies whether oil shocks bring relevant information once they pass through changes in financial conditions, represented here by the Treasury spread.

This question concerns not only researchers, but also investors, macro-financial monitoring and risk management. To an investor perspective, it is important to know how oil news should be interpreted directly as a bad news for equities or more indirectly like news about inflation or recession risk. For macro-financial monitoring, it helps to separate clearly between a commodity-market signal and a wider transmission process from oil shock to yield rates and finally to equities. For risk management, it matters because even modest predictive content can be useful if it helps identify periods in which macro shocks, curve repricing, and equity weakness become more closely connected. The purpose of an empirical analysis like this one is not to presume a forecasting gain but rather to see if the oil–yield curve–equity mechanism carries useful economic information.

1.3 Report Structure

The rest of this report is structured as follows. The next sections show the data construction, the sampling choice and how transformations are applied to oil, equity prices, and treasury yields. Section three contains descriptive statistics and diagnostics, including numerous tests of normality and independence. This is useful to characterize the important properties of the series before estimation. Section four is the core of the empirical analysis of this project. The project begins with an OLS regression, after which it extends the specification to a dynamic ARX(1) model that contains lagged S&P 500 returns, lagged oil and yield spread information. The report next studies the suggested transmission mechanism with the help of regressions. In support of these reduced form of regressions, a VAR(1) system is estimated for a consequent impulse-response analysis. The final section evaluates the forecasting performance through a rolling out-of-sample comparison across the models previously computed. To conclude, the report summarizes the main findings and discusses the limitations of empirical choices used for this report.

2 Literature Review

2.1 The Macroeconomic Impact of Oil Shocks

The relationship between oil price shocks and financial stability has been studied for decades. High increases in oil prices are usually viewed as a factor that precedes economic slowdowns. This concept has been studied by Hamilton (1983), who demonstrated that most of the U.S. recessions after World War II happened after a significant increase in crude oil prices. Hamilton showed that this correlation is statistically significant, suggesting that oil shocks were a contributing factor to these economic downturns. Then an oil price shock can be viewed as a factor that leads to declines in GNP growth.

However, while between 2003 and late 2007 there has been an unprecedented increase in oil prices, global financial markets and economies continued to grow. This apparent contradiction has been solved by showing that not all oil shocks are the same. Kilian (2009) established that the economic consequences of an oil shock depend on the cause of it. Kilian found three main drivers: disruptions to the physical supply of crude oil, shocks to demand for oil caused by changes in the business cycle, and oil-market specific demand shocks driven by shifting expectations.

It has been demonstrated that the U.S. stock market responds in highly different ways depending on the cause of the oil price fluctuation (Kilian & Park, 2009). For example, if oil prices increase because of a global demand boom, there is a positive effect of an expanding economy that outweighs the negative effect of higher costs. If prices spike because of an unexpected increase in demand caused by market fears over future supply shortage, the effect is negative and drives down equity prices. While physical supply disruptions have no significant effect on cumulative stock returns.

Since crude oil does not affect the economy in isolation, but its effects depend on the macroeconomic environment, modeling a direct, bivariate linear relationship between weekly WTI log returns and S&P 500 log returns often does not provide any significant result. To get a realistic picture of how an oil shock affects equities, the analysis needs to include an intermediate variable that can reflect changing financial and macroeconomic conditions.

2.2 Yield Spreads and Market Expectations

To understand these macroeconomic conditions, this project uses the spread between the 10-year and 2-year interest rates of the U.S. Treasury yield.

It is well documented that the shape of the yield curve is a reliable forecasting tool for U.S. recessions. The slope of the yield curve is based on the market's expectations on future inflation, monetary policies and overall growth (Estrella & Hardouvelis, 1991; Estrella & Mishkin, 1996). A steepening curve (widening spread) usually signals low probability of an economic downturn, while a flattening or inverted curve (shrinking or negative spread) anticipates a slowdown.

So, why is the spread important for understanding equity movements? Based on the premise that asset prices are determined by expectations of discounted future cash flows and discount rates, Campbell and Ammer (1993) analyzed how different types of "news" move both markets. Changes of the yield curve, that are caused by changes in the macroeconomic environment, modify the discount rates and growth forecasts used to price equities. However, the yield curve itself reacts to market expectations, therefore, it requires an initial macroeconomic shock to cause this mechanism.

2.3 Connecting Oil, Yields, and Equities

Bernanke et al. (1997) explain how a commodity shock causes this yield curve repricing. In their view, what really causes the economic damage when oil prices spike is rarely just the higher cost of energy itself, but the response of central banks to such a spike.

Since an oil price increase also generates a rise in inflation expectations, the Federal Reserve usually responds by tightening monetary policy and raising short-term interest rates. Meanwhile, long-term rates also rise by a smaller margin, reflecting the market's expectation that the stricter monetary policy will finally slow down the economy. Because short-term rates increase faster than long-term, an initial shock in oil price will cause the yield spread to flatten or invert.

It has also been analyzed how different types of oil price shocks affect the yield curve (Umar et al., 2022). The results confirm that the U.S. yield curve has a central role to transmit these shocks. The U.S. yield curve transmits the volatility caused by the oil shock across the global debt market, influencing the yield curves of other economies.

Therefore, looking at the oil-yield curve-equity mechanism helps better to explain oil shocks effects than looking at oil alone. By analyzing weekly changes in the yield spread together with oil returns, the analysis can understand how the market responds to the monetary policy changes as well as to the initial commodity shock. Since these variables continuously react to one another, it is important to determine the best models to capture these signals.

2.4 Forecasting Dynamics and Model Selection

It is difficult to translate this oil-yield curve-equity transmission mechanism into out-of-sample forecasting power. Classic financial literature demonstrates that equity returns are extremely noisy and that predictive models are unstable and fail to beat a simple unconditional benchmark, the historical average of the equity premium, in out-of-sample tests (Welch & Goyal, 2008).

However, simple univariate models ignore some aspects of the macroeconomic environment. For example, an AR(p) model only looks at the past returns of the S&P 500, not considering the signals coming from energy and bond markets. Since oil prices, yield curve and equities continuously interact and affect each other, multivariate models may be able to extract measurable predictive value.

Kang et al. (2014) applied Vector Autoregression (VAR) methods to model the interactions between oil price shocks and U.S. bond market returns. They used this multivariate approach to show the causes of oil price fluctuations and demonstrate the volatility effect between the markets. Their findings show that energy, debt, and equity markets are dynamically interdependent.

3 Data

3.1 Variables

To analyze the impact of an oil shock on the market, the chosen variables of interest are the West Texas Intermediate (WTI) Crude Oil spot price (oil_t), the S&P 500 equity index (spx_t), and the U.S. Treasury yield curve, represented by the 10-year ($r10_t$) and 2-year ($r2_t$) interest rates. The selection of these particular variables was driven by the necessity to capture the effect of the commodity shock in the supply (oil), the market response (U.S. equities), and the U.S. monetary policy (yield curve) and macroeconomic environment.

The period over which the analysis relies on spans from January 1990 to March 2026. The choice of this period make it possible to analyze over three decades of market cycles, monetary policies and substantially different macroeconomic shocks. Finally, the sample extends through the specific oil shock of February-March 2026, allowing to apply the tested models in a real-time forecasting scenario.

3.2 Data Transformations

The raw daily data was resampled to a weekly frequency (using the end-of-week (Friday) closing values) to capture the macroeconomic effect more effectively. Furthermore, following standard econometrics practices (magari reference?), the non-stationary price series were transformed into log returns. Let P_t denote the weekly price level; the continuous compounded return is defined as $r_t = \ln(P_t/P_{t-1})$ for both WTI crude (r_{oil_t}) and the S&P 500 (r_{spx_t}). To capture monetary policy response and expectations, we evaluate the yield curve spread as the difference between the 10-year and 2-year Treasury rates ($spread_t = r10_t - r2_t$), and compute its weekly first difference (d_spread_t).

3.3 Descriptive Statistics, Autocorrelation and Cross-Asset Dynamics

Table 1 presents the summary statistics for the transformed weekly variables. The data exhibit non-normality, Both the S&P 500 and WTI oil returns show negative skewness (-0.7899 and -0.6473, respectively), indicating fatter left tails. Finally, all three variables show high excess kurtosis, confirming the presence of fat tails, implying that extreme events (such as the 30% spike in WTI crude observed in 2026) happen more frequently than predicted by a Gaussian distribution. Furthermore, the Jarque-Bera test rejects the null hypothesis of normality for all series (p -value = 0.0000).

Finally, Autocorrelation Function (ACF) plots of the squared returns show significant volatility clustering. In addition, Ljung-Box tests confirm serial correlation across multiple lags in the series. Furthermore, an initial evaluation of the cross-asset correlation matrix shows that the correlation between weekly oil returns and the S&P 500 is relatively weak, showing that oil shocks do not instantly affect equity returns in a linear way.

Table 1: Summary Statistics for Weekly Variables

Variable	Mean	Std. Dev.	Skewness	Exc. Kurtosis
WTI Oil (Log Return)	0.0007	0.0502	-0.6473	7.0654
S&P 500 (Log Return)	0.0016	0.0232	-0.7899	6.9448
Spread Change (d_spread)	0.0002	0.0802	0.3041	3.1832

Note: Weekly resampled (Friday) observations. The Jarque-Bera test statistic is 4058.89 for Oil, 3990.42 for the S&P 500, and 826.22 for the Spread Change. The null hypothesis of normality is rejected at the 1% significance level for all variables.

4 Methodology

4.1 Empirical Strategy

The implemented empirical strategy aims to answer two questions. First, do oil price variations have intrinsic information for weekly S&P 500 returns if US Yield curve is taken into account? Second, are dynamic models better than static ones in out-of-sample forecasts relative to simpler benchmarks? This project wants to answer these questions by combining a single-equation approach with a multivariate time-series model, including a rolling forecasting exercise. This combination allows the analysis to document whether equity returns are linked to oil price shocks but also to prove if this information has practical and usable predictive value in the real world.

The project starts by taking daily observations on oil prices, the S&P 500 price index and finally the 10 year and 2 year US Treasury yield. This data sample is collected approximately from 1990 to March 2026. Using end of week observations, all series are transformed into weekly frequency. Using these weekly data, the analysis constructs oil returns $r_{oil,t} = \Delta \ln(oil_t)$, equity returns $r_{spx,t} = \Delta \ln(spx_t)$, the yield spread $spread_t = r_{10t} - r_{2t}$, and the weekly change in the spread $\Delta spread_t = spread_t - spread_{t-1}$. The use of weekly frequency seems an appropriate trade-off for the project's question rather than other time horizon. High frequency noises can come from daily data and reflects short term financial volatility, while monthly data would be too broad and may lose information.

The first stage of the analysis begins with descriptive statistics, normality tests, autocorrelation analysis and cross asset correlations in a way to characterize the variables' properties. This first step is crucial to evaluate whether the return series follow a Gaussian distribution, if dependence is strong over time or whether oil, bond and equity market move together in ways to support the project's proposed transmission mechanism. These results help determine which kind of regression to implement for the analysis.

As a benchmark, the project uses a static OLS regression where weekly S&P 500 returns are described by oil returns and changes in the yield spread, both with a weekly lag. With this logic the analysis implements a temporal ordering. This way, last week's information is used to forecast this week's stock return. This implementation does not aggregate a full causal identification because others relevant macroeconomic news or factors may still have an impact on what the project is studying.

The project then runs a residual diagnostics on the benchmark and this motivates the implementation of a dynamic specification. Auto correlation functions (ACF) plots and Ljung-Box tests applied to normal and squared residuals show that the static model does not take into account the commonly observed volatility clustering. Figure 1 shows the AFC of the OLS residuals and that the static regression leaves a non negligible dependence in the conditional mean, while figure 2 plots the AFC of the squared OLS, which shows the volatility clustering.

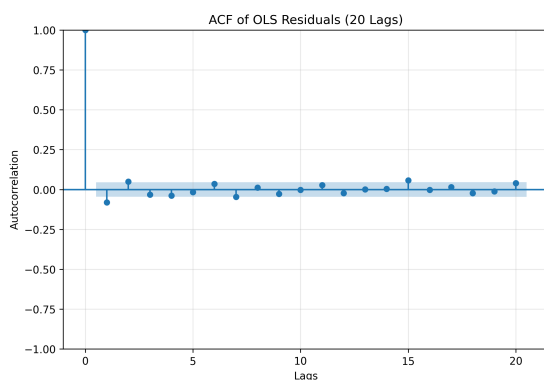


Figure 1: ACF of OLS residuals

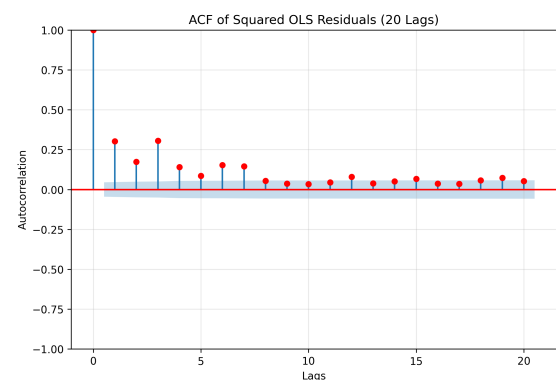


Figure 2: ACF of squared OLS residuals

To not exclude broader regressors, the analysis considers the ARMAX model and computes an appropriate specification. Selected candidates are then computed using autoregressive order $p \leq 3$ and moving-average order $q \leq 3$. Only models that converge and satisfy stability and invertibility are retained. Once estimated, the best model under the HQIN criteria is the ARMAX(1,0), equivalent to an ARX(1) model. The BIC criteria gives the same results. This then becomes the main equation used for the rest of the analysis.

The ARX(1) model, with its autoregressive component, carries a short run persistence, even said mean reversion in the equity market returns. The exogenous variables on the other hand stand for the two main component of this analysis, which are a direct oil to equity link and an indirect transmission through the bond market and expectations about the monetary policy. To better show this mechanism, the project complements a main return equation composed of two auxiliary regressions. The first one tries to understand whether oil price shocks can predict changes in the yield spread. The second one tests whether changes in the spread predict observed cumulative returns in the equity market over the next four weeks. With these regressions, the economic interpretation is more structured. If oil shocks have an influence over the yield curve and at the same time the change in spread predict future equity returns, then the yield curve may be interpreted as a channel linking energy markets and equities.

The analysis also estimates a VAR(1) for weekly oil returns and weekly changes in the spread and S&P 500 returns, which gives another multivariate benchmark where all variables move together. To conclude, then a rolling out of sample is computed to assesses predictive content. In order to do that, a 156 week rolling window is used, so that each week the models are re-estimated and can generate forecasts. This is done to compare the the ARX(1) and the VAR(1) model against the AR(1) and the random walk benchmark.

4.2 Model Specification

The first and main predictive regression of this analysis is:

$$r_{spx,t} = \alpha + \beta_1 r_{oil,t-1} + \beta_2 \Delta spread_{t-1} + \varepsilon_t \quad (1)$$

In this specification, $r_{oil,t-1}$ represents lagged oil shocks, and $\Delta spread_{t-1}$ stands for lagged variations in the yield-curve slope. The coefficient β_1 quantifies the direct forecasting power of oil-price fluctuations for equity market returns, while β_2 quantifies the influence of recent shifts in financial conditions and policy expectations.

For model selection, the project takes into consideration the more general ARMAX family:

$$r_{spx,t} = \alpha + \sum_{i=1}^p \phi_i r_{spx,t-i} + \beta_1 r_{oil,t-1} + \beta_2 \Delta spread_{t-1} + \varepsilon_t + \sum_{j=1}^q \theta_j \varepsilon_{t-j} \quad (2)$$

This regression allows weekly equity returns to depend on their own lags, on lagged oil shocks, on lagged changes in the yield spread, and eventually on serial dependence in the error term. As cited before, the analysis lead to implement an ARMAX(1,0). Since no moving average terms are included, the model can be written as an ARX(1):

$$r_{spx,t} = \alpha + \phi r_{spx,t-1} + \beta_1 r_{oil,t-1} + \beta_2 \Delta spread_{t-1} + \varepsilon_t. \quad (3)$$

The autoregressive term $\phi r_{spx,t-1}$ captures the short term existence or the weekly equity returns' mean reversion. The oil variable captures recent shocks in energy prices where the spread term links the monetary-policy and expectations channels through which changes in oil markets can influence equity prices.

Furthermore, the two transmission regressions are:

$$\Delta spread_t = \alpha + \gamma r_{oil,t-1} + u_t \quad (4)$$

and

$$\sum_{h=1}^4 r_{spx,t+h} = \alpha + \delta \Delta spread_t + v_t \quad (5)$$

Equation (4) is used to check if lagged oil shocks can predict changes in the yield curve. Equation (5) asks if today yield movements predict the equity performance over the following month.

The multivariate specification is a reduced VAR(1):

$$y_t = c + A_1 y_{t-1} + u_t \quad (6)$$

where y_t has weekly S&P 500 returns, weekly oil returns, and weekly changes in the yield spread. The VAR combines jointly all kinds of feedback effects across the three core variables of the analysis. This specification provides a better and broader understanding than a single equation model.

4.3 Estimation Procedure

After differencing, resampling and constructing the lag, the project estimates all models on aligned weekly samples. In this analysis, the OLS regression, ARX(1) model and the auxiliary regressions are estimated using the standard estimator form:

$$\hat{\beta} = (X'X)^{-1}X'y \quad (7)$$

Fitted values, residuals, standard errors, t -statistics, p -values, R^2 , adjusted R^2 , and the overall F -test are computed too to keep a transparent estimation of the various econometric values.

The project dynamic specification holds ARMAX(p, q) candidates that are estimated on a common sample and with comparable information criteria. The selection of model specification lead the analysis to use a ARMAX(1,0), equivalent to a ARX(1). As said before, residual diagnostics shows that the conditional mean is specified.

For VAR model, the analysis is conducted via the BIC, over a lag length from 1 to 12. The selected model is VAR(1) and its stability is verified thanks to the roots of its characteristic polynomial.

The last part of the estimation procedure is a forecasting exercise. The project uses a 156 week rolling window where at each time all models are newly calculated and a new *one-step-ahead* forecast for the weekly S&P 500 is computed. In this analysis, models compete among them. Their performance is then evaluated via a rolling week (52 week, 1 year) RMSE. The rolling coefficient of the ARX(1) is kept at each step of the computation to determine whether the oil to equity relationship is stable or it depends on time and across different market environments.

5 Results

5.1 Main Findings

The analysis begins with a simple OLS regression and then incorporates return dynamics using an ARX(1) specification. Both models are estimated on the same dataset of 1,887 weekly observations spanning from January 1990 to March 2026. This enables a direct comparison of how adding last week's equity return to the model changes the role of oil and the yield curve as predictors.

The static OLS model compares this week's S&P 500 log return, last week's oil log return and the the yield-curve spread.

$$r_{spx,t} = \alpha + \beta_1 r_{oil,t-1} + \beta_2 \Delta spread_{t-1} + \varepsilon_t \quad (1)$$

Table 2: Baseline OLS Regression Results

	Coefficient	Std. Error	<i>t</i> -statistic	<i>p</i> -value
Intercept	0.0016	0.0005	2.9954	0.0028
Lagged Oil Return	−0.0088	0.0107	−0.8243	0.4099
Lagged Spread Change	−0.0202	0.0067	−3.0062	0.0027
<i>n</i> = 1,887 <i>R</i> ² = 0.0056 Adj. <i>R</i> ² = 0.0045 <i>F</i> = 5.28 <i>p_F</i> = 0.0052				

The calculated coefficient on oil is negative but not statistically different from zero ($\hat{\beta}_1 = -0.0088$, $p = 0.41$), this implies that last week's change in oil prices has little value for forecasting equity returns. Alternatively, the spread in the yield curve is negative and significant at the 1% level ($\hat{\beta}_2 = -0.0202$, $p = 0.003$), this suggests that a yield curve flattening this week is related to lower equity returns next week. The regression is statistically significant ($F = 5.28$, $p = 0.005$), its explanatory power is extremely limited ($R^2 = 0.56\%$), in line with the widely recognized challenge of predicting short-horizon equity returns. (Welch & Goyal, 2008)

The residual diagnostics show clear limitations of the static model. As we can observe in Figure 3, the ACF of the squared residuals shows strong autocorrelation at every lag, confirming volatility clustering. Ljung-Box tests on the raw residuals also reject white noise at every lag tested, there is still predictable variation that the model fails to explain. These results make a strong case for moving to a dynamic specification.

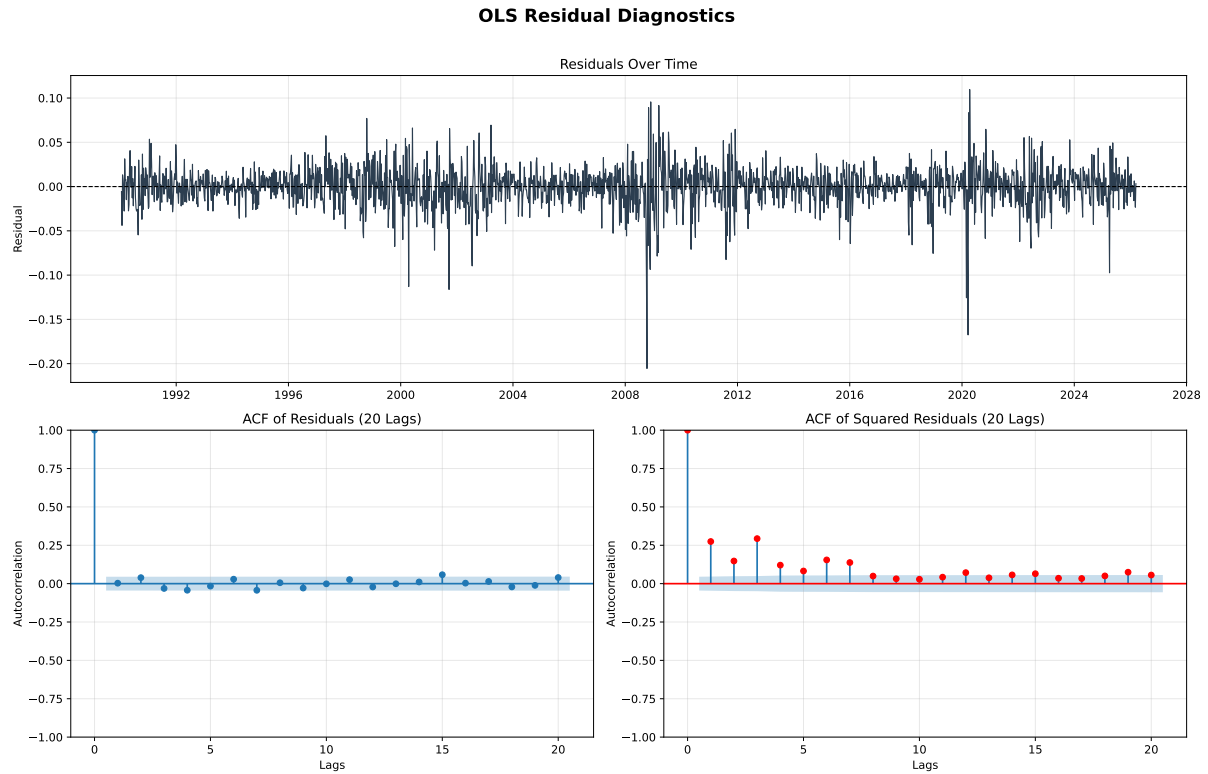


Figure 3: Top panel: Residuals over time. Bottom left: ACF of raw residuals. Bottom right: ACF of squared residuals. Strong autocorrelation in the series of squared values confirms volatility clustering. Spikes in the ACF indicate dependence in the conditional mean.

To fix the autocorrelation problem identified above, a lagged dependent variable is added. The lag order was selected through a grid search over all $\text{ARMAX}(p, q)$ specifications with $p, q \in \{0, 1, 2, 3\}$, keeping only models that satisfy stationarity and invertibility. The HQIC selects $\text{ARMAX}(1, 0)$ as the best model, which reduces to $\text{ARX}(1)$. The BIC criterion leads to the same result, confirming the robustness of this choice:

$$r_{spx,t} = \alpha + \phi r_{spx,t-1} + \beta_1 r_{oil,t-1} + \beta_2 \Delta spread_{t-1} + \varepsilon_t \quad (3)$$

Table 3: $\text{ARX}(1)$ OLS Estimation Results

	Coefficient	Std. Error	<i>t</i> -statistic	<i>p</i> -value
Intercept	0.0017	0.0005	3.2326	0.0012
Lagged SPX Return	-0.0834	0.0234	-3.5660	0.0004
Lagged Oil Return	-0.0010	0.0109	-0.0878	0.9301
Lagged Spread Change	-0.0204	0.0067	-3.0535	0.0023
<i>n</i> = 1,887 <i>R</i> ² = 0.0122 Adj. <i>R</i> ² = 0.0107 <i>F</i> = 7.78 <i>p_F</i> < 0.001				

First we notice that the $\text{AR}(1)$ term is negative and significant ($\hat{\phi} = -0.0834$, $p < 0.001$), proving evidence of weak short-term mean reversion in weekly equity returns. Second, after controlling for the autocorrelation in returns, the oil coefficient drops to near zero and becomes statistically meaningless ($\hat{\beta}_1 = -0.0010$, $p = 0.93$). This shows that oil was never really predicting equity returns, it was just filling in for the autocorrelation in returns that the simpler model had left out. Third, the yield curve coefficient remains negative and significant ($\hat{\beta}_2 = -0.0204$, $p = 0.002$), this confirms that the yield curve channel is robust to the inclusion of returns. The

overall fit improves moderately ($R^2 = 1.22\%$), and the F-test strongly rejects the null that all coefficients are jointly zero ($p < 0.001$).

Table 4: Static OLS vs ARX(1) — Coefficient Comparison

	Static OLS	ARX(1)	Difference	Std. Error (ARX)
Intercept	0.0016	0.0017	+0.0001	0.0005
Lagged SPX Return	—	−0.0834	—	0.0234
Lagged Oil Return	−0.0088	−0.0010	+0.0079	0.0109
Lagged Spread Change	−0.0202	−0.0204	−0.0003	0.0067

The spread coefficient is stable between the models (−0.0202 vs −0.0204), while the oil coefficient drops from −0.0088 to −0.0010, a fall of almost 90%. This supports the idea that the yield curve is the relevant channel connecting macro-financial conditions to equity returns, while the oil effect disappears once return dynamics are properly controlled for. The ARX(1) residuals no longer show significant autocorrelation, which means the AR(1) term has successfully absorbed the serial dependence found in the static baseline.

5.2 Forecasting Performance

The in-sample results give us evidence that the transmission chain from oil to equities through the yield curve is real. Statistical significance in-sample does not guarantee precise out-of-sample predictions. To assess this, we conducted a rolling forecasting exercise comparing four models over 1,731 weekly one-step-ahead forecasts from January 1993 until March 2026.

At each week t , all models are re-estimated using the past 156 weeks of data and a forecast is generated for the following week $t + 1$. Maintaining this order in time guarantees that no information from the future influences the analysis. The models we are comparing are the ARX(1), the VAR(1), the AR(1) as a univariate benchmark, and the Random Walk, which predicts zero at every step.

Table 5: Out-of-Sample RMSE Comparison — 4 Models

Model	RMSE	vs Random Walk
Random Walk	0.023680	baseline
AR(1)	0.023918	−1.01%
ARX(1)	0.024370	−2.92%
VAR(1)	0.024383	−2.97%

Based on 1,731 rolling weekly forecasts. Window size: 156 weeks.

Table 5 reports the RMSE for each model together with the comparison to the Random Walk percentage improvement or deterioration relative to the Random Walk baseline.

The Random Walk outperforms all structured models over the whole out-of-sample period. The AR(1) is closest to the Random Walk losing by only 1%, the ARX(1) and VAR(1) perform similarly, both around 3% worse than the Random Walk. The addition of oil and yield curve in the ARX(1) relative to the AR(1) does not improve forecasting accuracy, it actually worsens it.

Looking at the full-sample RMSE alone hides a lot of variations across different market regimes. Figure 4 shows the rolling 52-week RMSE for all four models from 1993 to 2026. During stable periods (1993–2007, 2012–2019), all four models track each other closely and the

Random Walk consistently outperforms. During crisis episodes the picture changes. The GFC 2008 pushes RMSE above 0.05. The COVID shock in 2020 generates a similar but shorter-lived spike. The Ukraine war and rate hiking cycle in 2022 produce a gradual increase. In all three episodes the four models remain clustered, with no single model gaining advantage over the others.

The graphs focusing on the crisis in Figure 4 confirm this pattern. During GFC 2008 the ARX(1) produces an average rolling RMSE of 0.03208 compared to 0.03102 for the Random Walk, the structured models perform slightly worse even during the oil-driven crisis, when they might be expected to perform best. During COVID 2020 and Ukraine 2022 the gap between the models is similarly small.

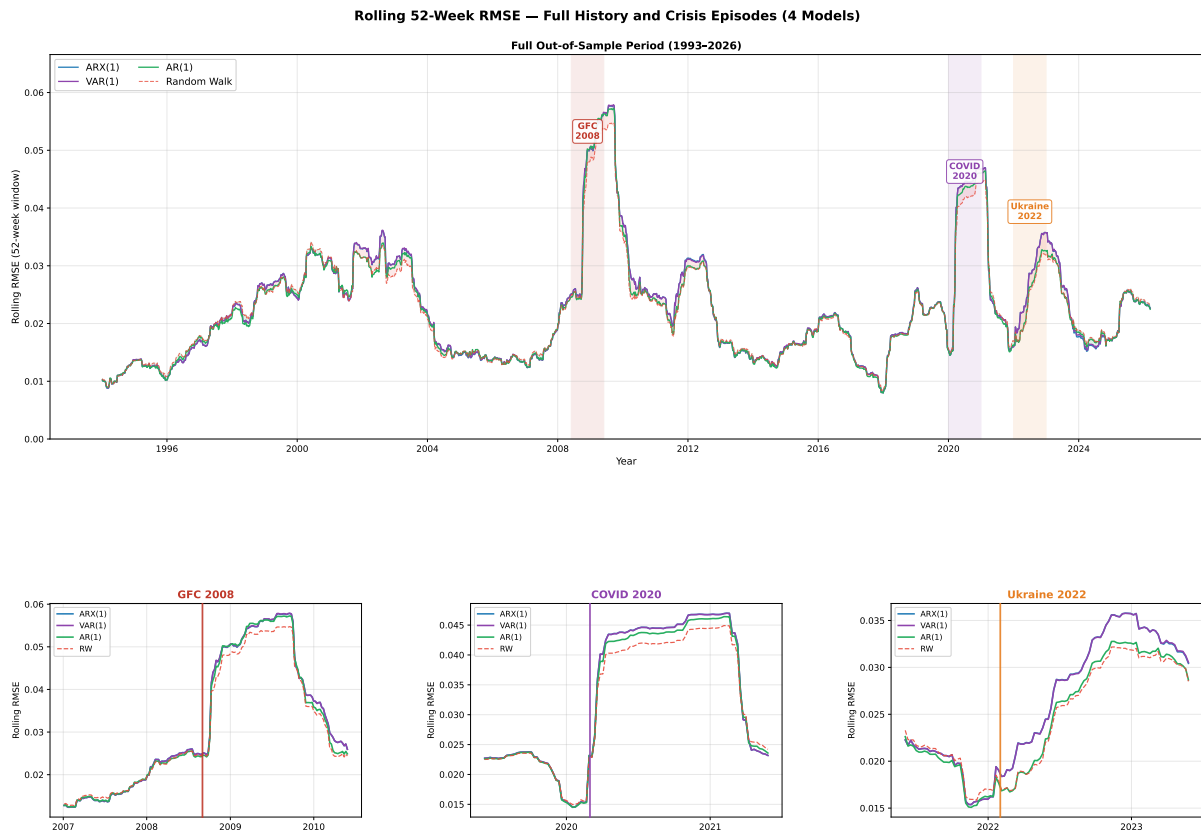


Figure 4: Rolling 52-week RMSE for all four models over the full out-of-sample period 1993–2026. The three bottom panels zoom in on the GFC 2008, COVID 2020 and Ukraine 2022.

Time-Varying Oil-Equity Relationship

Figure 5 gives a better insight on why the ARX(1) fails to beat the Random Walk despite in-sample significance of the yield curve channel. The figure plots the rolling estimate of $\hat{\beta}_{oil}$ which is the coefficient on lagged oil returns in the ARX(1), re-estimated weekly on a three-year window.

The full-sample estimate of $\hat{\beta}_{oil} = -0.0010$ is close to zero, this masks significant variation between regimes. During the global demand boom of 2004–2007, $\hat{\beta}_{oil}$ averaged -0.047 , meaning that oil increases were associated with falling equities, consistent with the interpretation of supply shock. During and after the GFC 2008, the coefficient shifted to positive (mean $+0.010$), reflecting the demand-driven nature of the oil price collapse and recovery. During the COVID and Ukraine period of 2020–2023, the coefficient averaged $+0.031$, positive as oil and equities moved together in the recovery phase before diverging sharply during the 2022 rate hikes.

This instability of the oil channel is the reason why we cannot generate accurate out-of-sample forecast. A model estimated on one regime produces a coefficient with the wrong sign

for the next regime and the average over the full sample converges to near zero. This finding is consistent with Kilian (2009), who shows that the effect of oil on macroeconomic variables depends on the nature of the shock, a distinction that a reduced-form linear model cannot capture.

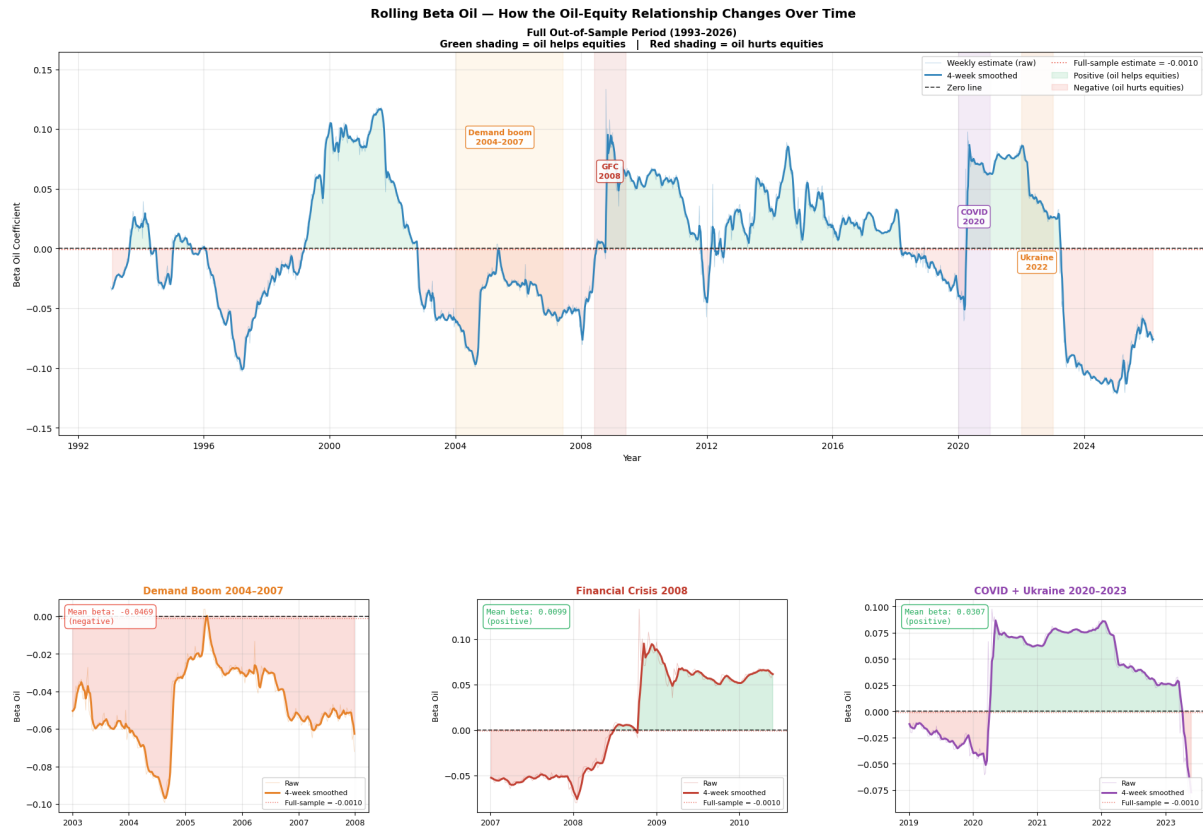


Figure 5: Rolling 3-year estimate of $\hat{\beta}_{oil}$ from the ARX(1) model. The blue line shows the 4-week smoothed series. Green shading indicates periods where oil helps equities (positive beta). Red shading indicates periods where oil hurts equities (negative beta). The bottom three panels zoom in on: demand boom 2004–2007, the GFC 2008 and the COVID–Ukraine 2020–2023 period.

5.3 Economic Interpretation

The results across Sections 5.1 and 5.2 paint a consistent and coherent picture. They suggest that the oil–yield curve–equity transmission channel exists and is directionally correct, but too weak and too regime-dependent to be exploited for short-horizon forecasting.

The Transmission Chain

The in-sample evidence supports the existence of the two-step mechanism described by Bernanke et al. (1997). Oil price increases are associated with yield curve flattening, yield curve flattening is associated with weaker forward equity returns. However, neither link is statistically significant on its own at weekly frequency, the VAR impulse response confirms that the equity reaction to an oil shock is never statistically different from zero at any point over the following six months. The channel exists in the data but operates too weakly to produce clean statistical evidence.

The comparison between static OLS and the ARX(1) add an important nuance. The direct oil-to-equity link that appears in the simpler model is not robust, it disappears once the dynamic structure of returns is controlled for. This suggests that oil does not affect equities directly in a linear and stable way, but rather through the macro-financial environment summarised by the yield curve. The yield curve coefficient, by contrast, survives the transition from static to

dynamic specification, confirming it as the more reliable of the two predictors.

Forecasting and Market Efficiency

The inability of any structured model to outperform the Random Walk in out-of-sample forecasting is consistent with the weak-form Efficient Market Hypothesis as described by Fama (1970). If oil returns and yield curve movements were consistently useful predictors, rational investors would have already incorporated this information into prices, eliminating the predictability. The fact that the Random Walk wins despite the statistical significance found in-sample is itself an informative result since it suggests that the in-sample relationships are too noisy and too unstable to survive genuine out-of-sample evaluation.

This result is also consistent with Welch and Goyal (2008), who document that macroeconomic predictors that seem significant in-sample, fail to beat simple benchmarks out-of-sample. Our analysis extends this finding to the case of oil returns and yield curve changes at weekly frequency.

Regime Dependence

The rolling beta analysis makes clear that the oil-equity relationship is fundamentally regime-dependent. The coefficient changes sign across different economic environments:

- Negative during demand-driven booms, when higher oil prices reflect global growth.
- Positive during supply-disruption crises, when oil and equities move together in the recovery phase.

This finding aligns directly with Kilian (2009), who demonstrates that macroeconomic consequences of oil shocks depend on their underlying source, and with Kilian and Park (2009), who document that the stock market reacts differently depending on whether the shock originates from supply or demand.

A linear model estimated over the full sample averages across these opposing regimes and produces a near-zero coefficient which provides no useful information for forecasting. This is not a failure of the empirical strategy, instead it is an economically meaningful result. It shows that the oil-equity relationship is not a stable structural feature of financial markets but rather a context-dependent phenomenon that requires knowledge of the nature of the shock to be interpretable.

6 Limitations

The results of this project show the extreme difficulty in an accurate forecasting of the equity markets out-of-sample. Indeed, as shown in the forecasting performance section, the simple Random Walk model is able to forecast with a lower RMSE than the dynamic and more complex models, with RMSE that were, for the ARX(1) and the VAR(1), 2.91% and 2.97% higher than the baseline benchmark, respectively.

This outcome confirms the Efficient Market Hypothesis (EMH): since markets are efficient, public information (such as oil price changes and yield curve movements) are immediately priced in stock prices. Then, evaluating real predictive power is difficult because shocks are priced in by the market instantly.

A second limitation regards the diagnostic analysis of the VAR model residuals. Indeed, the Ljung-Box tests show that the equations for oil and the yield spread still exhibit residual autocorrelation after three months. It is then evident that the model is not able to capture all the relevant aspects of those two variables.

Finally, one last limit is represented by the weekly data resampling choice. This choice was necessary for eliminating the noise of daily trading. But, it is probable that the stock market's reaction to oil shocks happens over just a few hours or days, then by aggregating the data on a weekly basis there is the risk of smoothing out the initial impact of the shock.

7 Conclusion

The project objective was to understand if it is possible to forecast U.S. stock market returns by looking at oil price shocks and taking into consideration the monetary policy response. The analysis did not just look at oil alone, but it also used the spread between the 10-year and 2-year Treasury yields as an intermediate transmission variable, testing the interaction among these factors through univariate and multivariate models (AR, Random Walk, ARX, and VAR).

The validity of the oil-yield curve-equity mechanism it has been studied and partially confirmed by the literature. A shock in oil price causes an increase in inflation, pushing the Federal Reserve to raise short-term rates (pushing up the 2-year yield), while investors prefer longer-term bonds fearing an economic slowdown (pushing down the 10-year yield). Therefore as the spread difference decreases stock prices incorporate the effects of higher discount rates.

However, going from theory to out-of-sample forecasting has shown significant limitations. The analysis and comparison of the forecasting power of univariate and multivariate models, showed that more complex and complete models (ARX and VAR) are able to capture some macroeconomic feedback effects, but fail to generate predictive advantage over a simple Random Walk. Finally, this project strongly supports the random walk hypothesis: oil shocks and interest rate fluctuations are priced in by the U.S. equity market too rapidly to be exploited by these models.

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